



CSCSV: Chinese Sentence Corpus with Semantic Variety

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Introduction

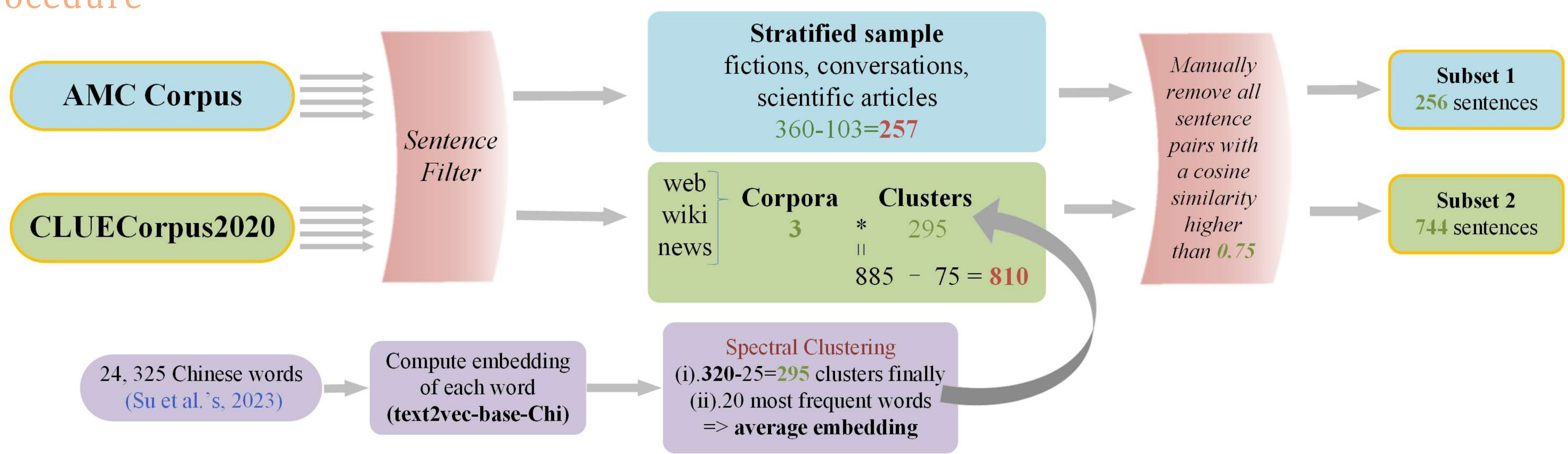
- In neuroimaging and behavioral research, the **precision and variety** of linguistic stimuli are critical for probing how the brain encodes and predicts meaning.
- However, existing Chinese sentence collections often either lack **fine-grained control** over **semantic similarity** or **cover only a narrow swath of conceptual space**—limitations that constrain the interpretability and generalizability of experimental findings.
- ❑ To address these gaps, we built the **Chinese Sentence Corpus with Semantic Variety (CSCSV)**, a set of **1,000** sentences whose lexical semantics are carefully balanced and whose pairwise similarities remain low (all cosine scores < **0.75**, mean $\approx$ **0.40**).

Significance

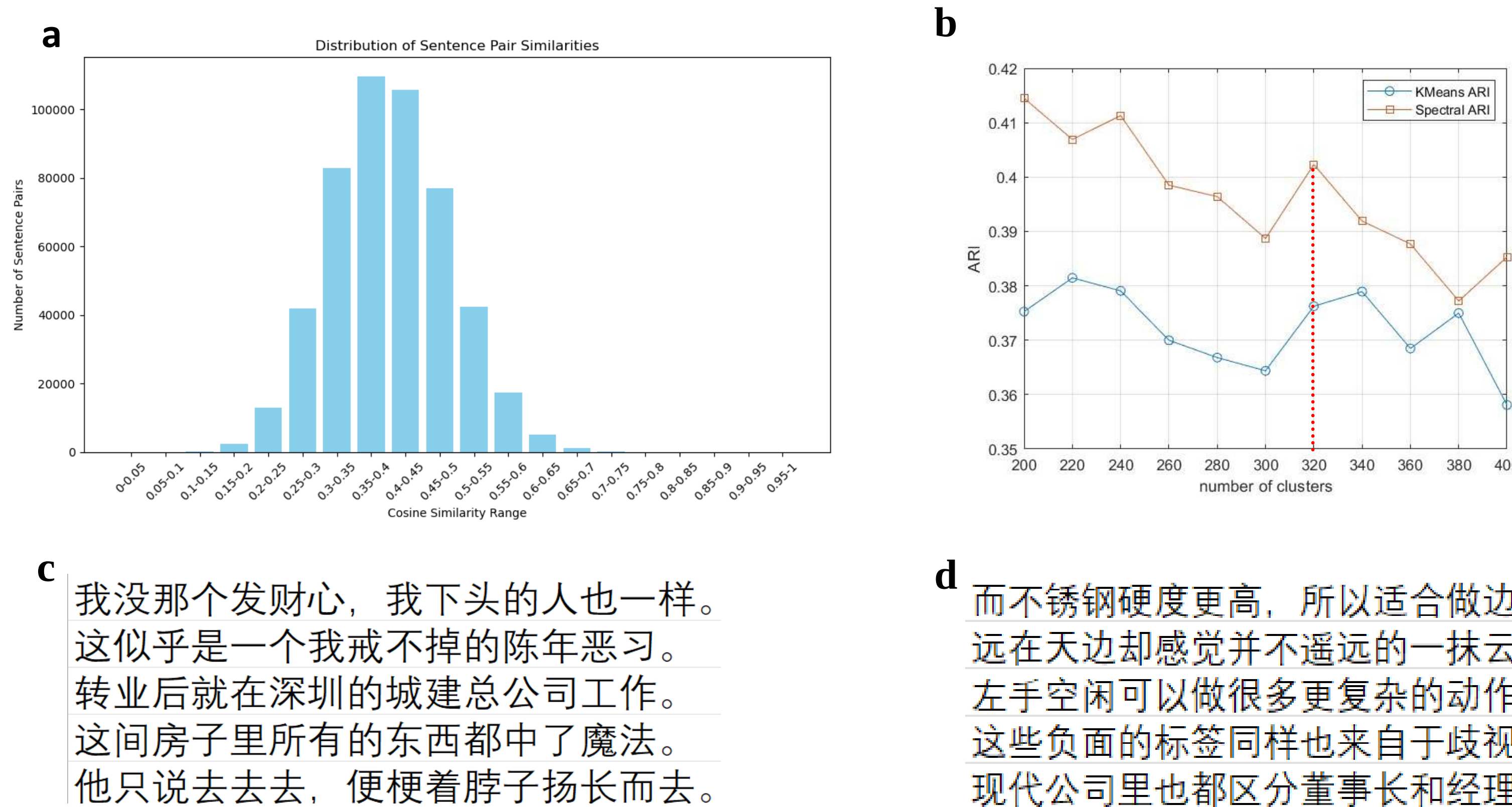
- ✓ Covering a broad spectrum of lexical-semantic categories, CSCSV offers standardized, semantically rich stimuli suitable for neuroimaging studies (e.g, fMRI, ECoG) focused on **predictive modelling**, as well as for behavioral experiments.

Materials and Methods

Procedure



Results & Quality Control

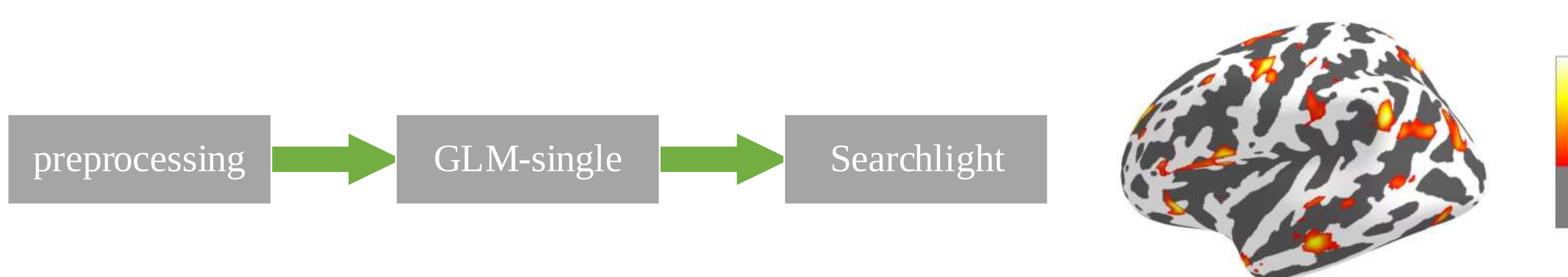


Sentence Filter Criteria

- Ensure each sentence has exactly **15** characters (excluding punctuation), counting digits and letters as one character each.
- Remove any **spaces** within the sentence and strip **newline characters** at the end.
- Ensure that no **non-Chinese** characters (e.g., Japanese characters) are present in the sentence.
- Exclude sentences that contain more than **5** (~66.67%) digits or English letters in total.
- Make sure each sentence ends with a period, exclamation mark, or question mark.

Ongoing work

- **Select Stimuli:** Identify sentences with extreme brain responses using an encoding model.
- **Measure Neural Activity:** Conduct fMRI scans with selected and baseline sentences.
- **Analyze Language Properties:** Link sentence features (e.g., syntax, surprisal) to neural responses.
- **Test LLM Control:** Validate if model-predicted stimuli causally modulate brain activity beyond natural language input.



Reference

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Contact

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